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ON MAHLER'S CONJECTURE FOR EVEN S-CONCAVE FUNCTIONS IN DIMENSIONS 1 AND 2.

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Abstract. In this paper, we establish different sharp forms of Mahler's conjecture for s-concave even functions in dimensions n, for $n = 1$ and 2, for $s > -1/n$, thus generalizing our previous results in [FN] on log-concave even functions in dimension 2, which corresponds to the case $s = 0$. The functional volume product of an even s-concave function g is

$$\int_{\mathbb{R}^n} g(x)dx \int_{\mathbb{R}^n} \mathcal{L}_s g(y)dy,$$

where $\mathcal{L}_s g$ is the s-polar function associated to g. The analogue of Mahler's conjecture for even s-concave functions postulates that this quantity is minimized for the indicatrix of a cube for any $s > -1/n$. In dimension $n = 1$, we prove this conjecture for all $s \in (-1, 0)$ (the case $s \geq 0$ was established by the first author and Mathieu Meyer in [FM10, page 17]). In dimension $n = 2$, we consider the case $1/s \in \mathbb{Z}$: for $s > 0$, we establish Mahler's conjecture for general s-concave even functions; for $s < 0$, we prove a sharp inequality for s-concave functions g such that g^s admits an asymptote in every direction. Notice that this set of functions is quite natural to consider, when $s < 0$, since it is the largest subset of s-concave functions stable by s-duality.

1. Introduction.

The classical Blaschke-Santaló inequality gives the following sharp relation between the volume of a centrally symmetric convex body K in $\mathbb{R}^n$ and the volume of its polar body K°:

$$P(K) = |K||K^\circ| \leq P(B_2^n),$$

where B_2^n is the Euclidean ball of radius one, $K^\circ = \{y \in \mathbb{R}^n, \langle x, y\rangle \leq 1, \forall x \in K\}$ is the polar body of K in $\mathbb{R}^n$ and $|A|$ stands for the Lebesgue measure of a Borel subset A of $\mathbb{R}^n$.

One of the well-known problems in the field of geometry of convex bodies is an inverse form of the Blaschke-Santaló inequality, namely Mahler's conjecture. This conjecture states that the product of the volume of a centrally symmetric convex body K and the volume of its polar body K°, should be minimized by the cube, that is,

$$P(K) \geq P([-1, 1]^n) = \frac{4^n}{n!}.$$

This conjecture was stated in 1938 by Mahler [Mah2]. The inequality in dimension two was established by Mahler himself [Mah1], the 3-dimensional inequality and the equality case were proved by Iriyeh and Shibata [IS], a shorter proof can be found in [FHMRZ]. The conjecture was also proved for several particular families of convex bodies like unconditional convex bodies by Saint Raymond [SR, M], zonoids by Reisner [Re, GMR], hyperplane sections of $B_p^n = \{x \in \mathbb{R}^n; \sum |x|_i^p \leq 1\}$ by Karasev [K].

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Functional forms of Mahler's conjecture have also been studied, where the centrally symmetric convex body is replaced with an even s-concave function and the polar body is replaced with the suitable s-polar of this function. More precisely, let us recall the definition of s-concavity.

Definition 1.1. Let $s \in \mathbb{R}$. A function $g : \mathbb{R}^n \to \mathbb{R}_+$ is s-concave, if for every $\lambda \in [0,1]$, and every x, $y \in \mathbb{R}^n$ such that $g(x)g(y) > 0$, one has

$$g((1-\lambda)x + \lambda y) \geq ((1-\lambda)g(x)^s + \lambda g(y)^s)^{\frac{1}{s}}, \quad \text{for } s \neq 0,$$

and $g((1-\lambda)x + \lambda y) \geq g(x)^{1-\lambda} g(y)^{\lambda}$, for $s = 0$.

Let $g : \mathbb{R}^n \to \mathbb{R}_+$ be an s-concave function not identically zero. Notice that for $s > 0$, it means that g^s is concave on its support, while for $s < 0$, it means that $g^s : \mathbb{R}^n \to \mathbb{R}_+ \cup \{+\infty\}$ is convex. For s-concave functions, the s-duality takes the following form. We define, for every $y \in \mathbb{R}^n$,

$$\mathcal{L}_s g(y) := \inf_{x \in \mathbb{R}^n} \frac{(1 - s\langle x, y\rangle)_+^{\frac{1}{s}}}{g(x)}, \quad \text{for } s \neq 0, \text{ and } \quad \mathcal{L}_0 g(y) = \inf_{x \in \mathbb{R}^n} \frac{e^{-\langle x,y\rangle}}{g(x)},$$

where the infimum are considered on the set $\{x \in \mathbb{R}^n; g(x) > 0\}$. Note that $\mathcal{L}_s g$ is s-concave and that $\mathcal{L}_0$ can also be described using the Legendre transform: for any function $\varphi : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$, one has $\mathcal{L}_0(e^{-\varphi}) = e^{-\varphi^*}$, where $\varphi^*(y) = \sup_x(\langle x, y\rangle - \varphi(x))$ denotes the Legendre transform of φ.

The following conjecture is the natural analogue of Mahler's conjecture for even s-concave functions.

Conjecture 1.2. Let $s > -\frac{1}{n}$ and $g : \mathbb{R}^n \to \mathbb{R}_+$ be an even s-concave function such that $0 < \int g < +\infty$. Then,

$$\int_{\mathbb{R}^n} g(x)dx \int_{\mathbb{R}^n} \mathcal{L}_s g(y)dy \geq \frac{4^n}{(1+s)\cdots(1+ns)},$$

with equality for $g = \mathbb{1}_{[-1,1]^n}$.

The conjecture has been proved for $s = 0$ and g being unconditional in [FM08a, FM08b]. Recall that a function is unconditional if $g(|x_1|, ..., |x_n|) = g(x_1, ..., x_n)$ for every $(x_1, ..., x_n) \in \mathbb{R}^n$. It has also been established for $s > 0$ and $n = 1$ in [FM10, page 17]; for $s > 0$ and g unconditional in [BF, FGSZ] and for $s = 0$ and $n = 2$ in [FN].

According to the sign of s, there are two ways of reformulating Conjecture 1.2. If $s > 0$, then it is equivalent to finding the exact lower bound for the quantity

$$\int_{\mathbb{R}^n} f(x)^m dx \int_{\mathbb{R}^n} (\mathcal{L}_1 f(y))^m dy,$$

where f is an even concave function such that $0 < \int f^m < +\infty$. While for $s < 0$, Conjecture 1.2 is equivalent to finding the exact lower bound for

$$\int_{\mathbb{R}^n} \frac{1}{f(x)^m} dx \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}f(y))^m} dy,$$

where $f : \mathbb{R}^n \to (0, +\infty]$ is even, convex, f goes to infinity at infinity and for every $y \in \mathbb{R}^n$

$$\mathcal{M}f(y) = \sup_{x \in \mathbb{R}^n} \frac{1 + \langle x, y\rangle}{f(x)}.$$

In our first main result, we prove Conjecture 1.2 in the case where $1/s \in \mathbb{N}$, and in dimension $n = 2$.

Theorem 1.3. *Let $s > 0$ be such that $1/s \in \mathbb{N}$ and let $g : \mathbb{R}^2 \to \mathbb{R}_+$ be an even s-concave function such that $0 < \int g < +\infty$. Then,*

$$\int_{\mathbb{R}^2} g(x)dx \int_{\mathbb{R}^2} \mathcal{L}_s g(y)dy \geq \frac{16}{(1+s)(1+2s)},$$

with equality for $g = \mathbb{1}_{[-1,1]^2}$.

Notice that letting $s \to 0$, we get a new proof of the log-concave case from [FN], though without the equality case, see Corollary 3.3.

For $s < 0$, the situation is more involved, because the s-dual $\mathcal{L}_s g$ of an integrable s-concave function g is not only s-concave but, as observed in [Ro, FGSZ], the function $(\mathcal{L}_s g)^s$ has also asymptotes in every direction. For example, for $g = \mathbb{1}_{[-1,1]^n}$, one has $(\mathcal{L}_s g(y))^s = 1 - s\|y\|_1$, where $\|y\|_1 = \sum_{i=1}^n |y_i|$, for $y = (y_1, \dots, y_n)$. More precisely, in general, one has $(\mathcal{L}_s g)^s \in \mathcal{F}$, where the set $\mathcal{F}$ is defined below.

Definition 1.4. $\mathcal{F}$ is the set of convex lower semi-continuous functions $f : \mathbb{R}^n \to (0, +\infty)$ such that, for any $x \neq 0$, one has $\lim_{t\to+\infty} f(tx) = +\infty$, and $t \mapsto \frac{f(tx)}{t}$ is non-increasing on $(0, +\infty)$.

Notice that if $g^s \in \mathcal{F}$, then it is not difficult to see that $0 < \int g < +\infty$, see the inequality (2.3) in section 2. Moreover, it is easy to see that for any g such that $g^s \in \mathcal{F}$, one has $\mathcal{L}_s\mathcal{L}_s g = g$, from which it follows that for any map g, one has $\mathcal{L}_s\mathcal{L}_s\mathcal{L}_s g = \mathcal{L}_s g$. As we shall see from the one-dimensional case, it is natural to formulate two more conjectures from which Conjecture 1.2 follows in the case $s < 0$. The first one postulates a sharp lower bound on the functional volume product of s-concave even functions, among the ones such that $g^s \in \mathcal{F}$.

Conjecture 1.5. Let $s \in (-\frac{1}{n}, 0)$ and let $g : \mathbb{R}^n \to \mathbb{R}_+$ be even such that $g^s \in \mathcal{F}$. Then,

$$\int_{\mathbb{R}^n} g(x)dx \int_{\mathbb{R}^n} \mathcal{L}_s g(y)dy \geq \frac{1}{(1+ns)} \times \frac{4^n}{(1+s)\cdots(1+ns)},$$

with equality for $g(x) = (\max(1, \|x\|_\infty))^{1/s}$.

Notice that $(\max(1, \|x\|_\infty))^{1/s} = \mathcal{L}_s\mathcal{L}_s(\mathbb{1}_{[-1,1]^n})$. In [FGSZ], the above conjecture was established if the function g is unconditional. In particular, the conjecture holds for $n = 1$. In our second main result, we establish Conjecture 1.5 when $n = 2$ and $-1/s \in \mathbb{N}$.

Theorem 1.6. *Let $s \in (-\frac{1}{2}, 0)$ be such that $-1/s \in \mathbb{N}$ and let $g : \mathbb{R}^2 \to \mathbb{R}_+$ be even such that $g^s \in \mathcal{F}$. Then,*

$$\int_{\mathbb{R}^2} g(x)dx \int_{\mathbb{R}^2} \mathcal{L}_s g(y)dy \geq \frac{16}{(1+s)(1+2s)^2},$$

with equality for $g(x) = (\max(1, \|x\|_\infty))^{1/s}$.

The second conjecture compares the integral of g with the integral of its double s-polar.

Conjecture 1.7. Let $s \in (-\frac{1}{n}, 0)$ and let $g : \mathbb{R}^n \to \mathbb{R}_+$ be even s-concave such that $0 < \int g < +\infty$. Then,

$$\int_{\mathbb{R}^n} g(x)dx \geq (1+ns) \int_{\mathbb{R}^n} \mathcal{L}_s\mathcal{L}_s g(x)dx, \tag{1.1}$$

with equality for $g = \mathbb{1}_{[-1,1]^n}$.

Notice that, for any function $g : \mathbb{R}^n \to \mathbb{R}_+$, one has, for any $x \in \mathbb{R}^n$, $g(x) \leq \mathcal{L}_s\mathcal{L}_s g(x)$ thus $\int_{\mathbb{R}^n} g(x)dx \leq \int_{\mathbb{R}^n} \mathcal{L}_s\mathcal{L}_s g(x)dx$. The conjectured inequality (1.1) postulates that one has an opposite bound up to constant for s-concave even functions.

In our third main result, we establish Conjecture 1.7 in dimension $n = 1$.

Theorem 1.8. *Let $s \in (-1, 0)$ and let $g : \mathbb{R} \to \mathbb{R}_+$ be even s-concave such that $0 < \int g < +\infty$. Then,*

$$\int_{\mathbb{R}} g(x)dx \geq (1+s) \int_{\mathbb{R}} \mathcal{L}_s\mathcal{L}_s g(y)dy,$$

with equality for $g = \mathbb{1}_{[-1,1]}$.

Remark 1.9. Notice that, if Conjecture 1.7 holds for any s-concave even function g and Conjecture 1.5 holds for $\mathcal{L}_s g$, then multiplying equation (1.1) by $\int \mathcal{L}_s g$, using that $(\mathcal{L}_s g)^s \in \mathcal{F}$ and applying Conjecture 1.5 to $\mathcal{L}_s g$, then we deduce that Conjecture 1.2 holds for g.

From the above remark, the results in [FGSZ] and Theorem 1.8, one deduces that Conjecture 1.2 holds for $n = 1$ and $s \in (-1, 0)$, which is the following theorem.

Theorem 1.10. *Let $s \in (-1, 0)$ and $g : \mathbb{R} \to \mathbb{R}_+$ be even convex such that $0 < \int g < +\infty$. Then,*

$$\int_{\mathbb{R}} g(x)dx \int_{\mathbb{R}} \mathcal{L}_s g(y)dy \geq \frac{4}{(1+s)^2},$$

with equality if and only if $g = c\mathbb{1}_{[-a,a]}$, for some $c > 0$ and $a > 0$.

The proof's strategy of our main theorems in dimension $n = 2$ relies on constructing an associated symmetric convex body in dimension $m + 2$, where $m = 1/|s|$, to which we apply the following proposition concerning Mahler's conjecture in higher dimensions proved in [FHMRZ].

Proposition 1.11. [FHMRZ] *Let $K \subset \mathbb{R}^n$ be a centrally symmetric convex body that can be partitioned with hyperplanes H_1, H_2,..., H_n into 2^n pieces of the same volume such that each section $K \cap H_i$ satisfies Mahler's conjecture and is partitioned into 2^{n-1} regions of the same $(n-1)$-dimensional volume by the remaining hyperplanes, then*

$$P(K) \geq \frac{4^n}{n!}.$$

In our proofs, we shall need the following notion of equipartition of functions.

Definition 1.12. Let $m \in \mathbb{Z}$. A function $f : \mathbb{R}^n \to \mathbb{R}_+$ is said to be m-equipartitioned if

$$\int_{\mathbb{R}^n_\varepsilon} f^m = \frac{1}{2^n} \int_{\mathbb{R}^n} f^m \quad \text{and} \quad \int_{\mathbb{R}^n_\varepsilon} f^{m-1} = \frac{1}{2^n} \int_{\mathbb{R}^n} f^{m-1},$$

where $\forall \varepsilon \in \{-1, 1\}^n$, $\mathbb{R}^n_\varepsilon = \{x \in \mathbb{R}^n; \varepsilon_i x_i \geq 0, \forall i \in \{1, ..., n\}\}$.

This paper is organized in the following way: In section 2, we present some general results and properties of the s-dual function $\mathcal{L}_s$ and we introduce the construction of convex bodies using s-concave functions. In section 3, we establish Conjecture 1.2 in dimensions $n = 1$ and $n = 2$ for $\frac{1}{s} \in \mathbb{N}$. In section 4, we give a lower bound of the functional volume product applied to even s-concave functions in dimensions $n = 1$ and $n = 2$ for $-\frac{1}{s} \in \mathbb{N}$.

2. Construction and properties of associated convex bodies.

Let $s > -\frac{1}{n}$ and $g : \mathbb{R}^n \to \mathbb{R}_+$ be an s-concave function. We discuss two cases according to the sign of s.

2.1. **Case $s > 0$.** Let $m = \frac{1}{s} > 0$. Assume that $0 < \int g < +\infty$. Then there exists a function $f : \mathbb{R}^n \to \mathbb{R}_+$ concave on its support such that $g = f^{\frac{1}{s}} = f^m$ and

$$\mathcal{L}_s g(y) = \mathcal{L}_s f^{\frac{1}{s}}(y) = \inf_{x \in \mathbb{R}^n} \frac{(1 - s\langle x, y\rangle)_+^{\frac{1}{s}}}{f(x)^{\frac{1}{s}}} = \left(\inf_{x \in \mathbb{R}^n} \frac{(1 - \langle x, sy\rangle)_+}{f(x)} \right)^{\frac{1}{s}} = (\mathcal{L}f(sy))^{\frac{1}{s}} = \left(\mathcal{L}f\left(\frac{y}{m}\right)\right)^m,$$

where $\mathcal{L}f(y) = \mathcal{L}_1 f(y) = \inf_{x \in \mathbb{R}^n} \frac{(1-\langle x,y\rangle)_+}{f(x)}$. In addition, by the change of variables $y = mz$, we get

$$\int_{\mathbb{R}^n} \mathcal{L}_s g(y)dy = m^n \int_{\mathbb{R}^n} (\mathcal{L}f(z))^m dz.$$

Introducing the operator P_m defined as follows

$$P_m(f) = \int_{\mathbb{R}^n} f(x)^m dx \int_{\mathbb{R}^n} (\mathcal{L}f(y))^m dy,$$

and changing variables, we get

$$\int_{\mathbb{R}^n} g(x)dx \int_{\mathbb{R}^n} \mathcal{L}_s g(y)dy = m^n P_m(f).$$

Thus, for $s > 0$, Conjecture 1.2 can equivalently be formulated in the following form: for any even function $f : \mathbb{R}^n \to \mathbb{R}_+$, concave on its support and integrable and for any $m > 0$, one should have

$$P_m(f) = \int_{\mathbb{R}^n} f(x)^m dx \int_{\mathbb{R}^n} (\mathcal{L}f(y))^m dy \geq P_m(\mathbb{1}_{[-1,1]^n}) = \frac{4^n}{(m+1)\cdots(m+n)}. \tag{2.1}$$

The operator P_m is invariant under linear transformation. Indeed, let $T : \mathbb{R}^n \to \mathbb{R}^n$ be an invertible linear map. Then, putting $z = Tx$ we get,

$$\int_{\mathbb{R}^n} (f \circ T(x))^m dx = \frac{1}{|\det T|} \int_{\mathbb{R}^n} f(z)^m dz.$$

We also obtain, for every $y \in \mathbb{R}^n$,

$$\mathcal{L}(f \circ T)(y) = \inf_{x \in \mathbb{R}^n} \frac{(1 - \langle x, y\rangle)_+}{f(Tx)} = \inf_{z \in \mathbb{R}^n} \frac{\left(1 - \langle T^{-1}z, y\rangle\right)_+}{f(z)} = \mathcal{L}f\left((T^{-1})^* y\right).$$

Therefore, $\mathcal{L}(f \circ T) = (\mathcal{L}f) \circ (T^{-1})^*$. Hence, changing variables, we get

$$\begin{aligned} P_m(f \circ T) &= \int_{\mathbb{R}^n} (f(Tx))^m dx \int_{\mathbb{R}^n} (\mathcal{L}f((T^{-1})^* y))^m dy \\ &= \frac{1}{|\det T|} \int_{\mathbb{R}^n} f(x)^m dx \times |\det T^*| \int_{\mathbb{R}^n} (\mathcal{L}f(y))^m dy = P_m(f). \end{aligned}$$

Let us introduce some tools that will help us in the proof of our main theorem in section 3.
Let L be a symmetric convex body in $\mathbb{R}^m$. Define the set

$$K_{L,m}(f) = \{(x,t) \in \mathbb{R}^n \times \mathbb{R}^m; \|t\|_L \leq f(x)\},$$

where $\|.\|_L$ is defined for $y \in \mathbb{R}^m$ by $\|y\|_L = \inf\{\lambda > 0; y \in \lambda L\}$. Variants of these sets and their volume products were considered in [AKM, BF] but we establish their properties below for completeness. Since f is even and concave, then $K_{L,m}(f)$ is a symmetric convex body in $\mathbb{R}^{m+n}$ and for every $m > 0$ one has

$$|K_{L,m}(f)| = |\{(x,t) \in \mathbb{R}^n \times \mathbb{R}^m; \|t\|_L \leq f(x)\}| = \int_{\mathbb{R}^n} |f(x)L|_m dx = |L| \int_{\mathbb{R}^n} f(x)^m dx.$$

Moreover, we have

$$\begin{aligned} (K_{L,m}(f))^\circ &= \{(y,s) \in \mathbb{R}^n \times \mathbb{R}^m; \langle x, y\rangle + \langle s, t\rangle \leq 1, \quad \forall (x,t) \in K_{L,m}(f)\} \\ &= \{(y,s) \in \mathbb{R}^n \times \mathbb{R}^m; \sup_{\{t \in \mathbb{R}^m; \|t\|_L \leq f(x)\}} \langle s, t\rangle \leq 1 - \langle x, y\rangle, \quad \forall x \in \mathbb{R}^n\} \\ &= \{(y,s) \in \mathbb{R}^n \times \mathbb{R}^m; f(x)\|s\|_{L^\circ} \leq (1 - \langle x, y\rangle)_+, \quad \forall x \in \mathbb{R}^n\} \\ &= \{(y,s) \in \mathbb{R}^n \times \mathbb{R}^m; \|s\|_{L^\circ} \leq \mathcal{L}f(y)\} \\ &= K_{L^\circ,m}(\mathcal{L}f). \end{aligned}$$

Hence

$$|K_{L,m}(f)^\circ| = |K_{L^\circ,m}(\mathcal{L}f)| = |L^\circ| \int_{\mathbb{R}^n} (\mathcal{L}f(y))^m dy.$$

Thus, one has

$$P(K_{L,m}(f)) = |K_{L,m}(f)||K_{L^\circ,m}(\mathcal{L}f)| = |L||L^\circ| \int_{\mathbb{R}^n} f(x)^m dx \int_{\mathbb{R}^n} (\mathcal{L}f(y))^m dy = P(L)P_m(f).$$

We need the following lemma describing the relation between the concave function f and its dual.

Lemma 2.1. *Let $f:\mathbb{R}^n\to\mathbb{R}_+$ be a concave function on its support. Then,*

$$\mathrm{supp}(\mathcal{L}f)=(\mathrm{supp}f)^\circ.$$

Proof. Let $C=\mathrm{supp}(f)$ and $K(f)=\{(x,t)\in C\times\mathbb{R};|t|\le f(x)\}$. We know that $K(f)$ is convex in $\mathbb{R}^{n+1}$ and $K(f)^\circ=K(\mathcal{L}f)$. Since the operation of duality transforms sections of a given convex body into projections of its polar and

$$K(f)\cap e_{n+1}^\perp=C=P_{e_{n+1}^\perp}(K(f)),$$

we have

$$K(\mathcal{L}f)\cap e_{n+1}^\perp=K(f)^\circ\cap e_{n+1}^\perp=\left(P_{e_{n+1}^\perp}(K(f))\right)^\circ.$$

Thus,

$$\mathrm{supp}\mathcal{L}f=C^\circ=(\mathrm{supp}f)^\circ.$$

□

2.2. **Case $-\frac{1}{n}<s<0$.** Let $m=-\frac{1}{s}>n$. Then there exists a convex function $f:\mathbb{R}^n\to(0,+\infty]$ such that $f=g^s$. Assume that $0<\int g<+\infty$, one has $\lim_{t\to+\infty}f(tx)=+\infty$, for any $x\neq 0$. Moreover, since $s=-1/m$ and $g=f^{-m}$,

$$\mathcal{L}_sg(y)=\inf_{x\in\mathbb{R}^n}\frac{(1-s\langle x,y\rangle)_+^{\frac{1}{s}}}{g(x)}=\left(\inf_{x\in\mathbb{R}^n}\frac{(1+\langle x,\frac{y}{m}\rangle)_+^{-1}}{f^{-1}(x)}\right)^m=\left(\mathcal{L}_{-1}(f^{-1})\left(\frac{y}{m}\right)\right)^m.$$

Let us define the function $\mathcal{M}f$, for $y\in\mathbb{R}^n$, by

$$\mathcal{M}f(y)=(\mathcal{L}_{-1}(f^{-1})(y))^{-1}=\sup_{x\in\mathbb{R}^n}\frac{1+\langle x,y\rangle}{f(x)}.$$

It was proved in [FGSZ] that, with these hypotheses, the function $\mathcal{M}f$ belongs to $\mathcal{F}$ (see definition 1.4). Using the change of variables $y=mz$, we get

$$\int_{\mathbb{R}^n}\mathcal{L}_sg(y)dy=m^n\int_{\mathbb{R}^n}\left(\mathcal{L}_{-1}(f^{-1})(z)\right)^m dz=m^n\int_{\mathbb{R}^n}\frac{1}{(\mathcal{M}f(z))^m}dz.$$

Consider the operator Q_m defined as follows

$$Q_m(f)=\int_{\mathbb{R}^n}\frac{1}{f(x)^m}dx\int_{\mathbb{R}^n}\frac{1}{(\mathcal{M}f(y))^m}dy,$$

one has

$$\int_{\mathbb{R}^n}g(x)dx\int_{\mathbb{R}^n}\mathcal{L}_sg(y)dy=m^n\int_{\mathbb{R}^m}\frac{1}{f(x)^m}dx\int_{\mathbb{R}^n}\frac{1}{(\mathcal{M}f(z))^m}dz=m^nQ_m(f).$$

Thus, for $s<0$, Conjecture 1.2 can equivalently be formulated in the following form: for any even function $f:\mathbb{R}^n\to(0,+\infty]$, convex, such that $\lim_{t\to+\infty}f(tx)=+\infty$, for any $x\neq 0$ and for any $m>n$, one should have

$$Q_m(f)=\int_{\mathbb{R}^n}\frac{1}{f(x)^m}dx\int_{\mathbb{R}^n}\frac{1}{(\mathcal{M}f(y))^m}dy\ge Q_m(1+I_{[-1,1]^n})=\frac{4^n}{(m-1)\cdots(m-n)},\tag{2.2}$$

where, for any $K\subset\mathbb{R}^n$, the function I_K is defined by $I_K(x)=0$ for $x\in K$ and $I_K(x)=+\infty$ for $x\notin K$.

On the other hand, if $f\in\mathcal{F}$, then it was proved in [FGSZ] that for any $u\in S^{n-1}$ there exists a continuous positive function a such that $f(tu)\ge a(u)t$ for every $t>0$. By compactness of the sphere and continuity of a there exists $c>0$ such that $f(tu)\ge ct$ for every $t>0$. We deduce that $f(x)\ge c\|x\|_2$ for every $x\in\mathbb{R}^n$. Moreover, since f is positive, continuous and goes

to infinity at infinity, there exists $c' > 0$ such that $f(x) \geq c'$ for every $x \in \mathbb{R}^n$. We conclude that $f(x) \geq \max(c\|x\|_2, c')$. Thus,

$$g(x) = f(x)^{\frac{1}{s}} \leq \max(c\|x\|_2, c')^{\frac{1}{s}}, \tag{2.3}$$

which is integrable since $-\frac{1}{n} < s < 0$. Therefore, for any $f \in \mathcal{F}$, the function $g = f^{\frac{1}{s}}$ is integrable.

Same as before, Conjecture 1.5 can be reformulated in the form: for $f \in \mathcal{F}$ even and $m > n$,

$$Q_m(f) \geq Q_m(\max(1, \|x\|_\infty)) = \frac{m}{m-n} \times \frac{4^n}{(m-1)\cdots(m-n)}. \tag{2.4}$$

And Conjecture 1.7 reads as: for any even function $f : \mathbb{R}^n \to (0, +\infty]$, convex such that $\lim_{t\to+\infty} f(tx) = +\infty$, for any $x \neq 0$ and for any $m > n$,

$$\int_{\mathbb{R}^n} \frac{1}{f(x)^m} dx \geq \frac{m-n}{m} \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}\mathcal{M}f(x))^m} dx, \tag{2.5}$$

with equality for $f = 1 + I_{[-1,1]^n}$.

Similarly as before, one may easily prove that the operator Q_m is invariant under linear transformation. Let $K \subset \mathbb{R}^m$ be a symmetric convex body. For any even convex function $f : \mathbb{R}^n \to \mathbb{R}_+$ such that $\lim_{x\to\infty} f(tx) = +\infty$, for any $x \neq 0$, we define the function

$$\begin{array}{rcl} \pi_m(f) \ : \ \mathbb{R}^n \times \mathbb{R}^m & \to & \mathbb{R}_+ \cup \{+\infty\} \\ (x,s) & \mapsto & \pi_m(f)(x,s) = \begin{cases} \|s\|_K f(\frac{x}{\|s\|_K}) & \text{if } s \neq 0. \\ \lim_{s\to 0} \|s\|_K f(\frac{x}{\|s\|_K}) & \text{if } s = 0. \end{cases} \end{array}$$

We denote the domain of f by $\mathrm{dom}(f) = \{x \in \mathbb{R}^n; f(x) < +\infty\}$. Then

$$\mathrm{dom}(\pi_m(f)) = \{(x,s) \in \mathbb{R}^n \times \mathbb{R}^m; \pi_m(f)(x,s) < +\infty\} = \{(x,s) \in \mathbb{R}^n \times \mathbb{R}^m; x \in \|s\|_K \mathrm{dom}(f)\}.$$

Consider the following convex body

$$C_m^K(f) = \{(x,s) \in \mathbb{R}^n \times \mathbb{R}^m; \pi_m(f)(x,s) \leq 1\} = \overline{\left\{(x,s) \in \mathbb{R}^n \times \mathbb{R}^m \setminus \{0\}; \|s\|_K f\left(\frac{x}{\|s\|_K}\right) \leq 1\right\}}.$$

One has,

$$\begin{aligned} (C_m^K(f))^\circ &= \{(y,t) \in \mathbb{R}^n \times \mathbb{R}^m; \langle (x,s),(y,t)\rangle \leq 1, \forall (x,s) \in C_m^K(f)\} \\ &= \left\{(y,t) \in \mathbb{R}^n \times \mathbb{R}^m; \langle x,y\rangle + \langle s,t\rangle \leq 1, \forall (x,s); \|s\|_K f\left(\frac{x}{\|s\|_K}\right) \leq 1\right\}. \end{aligned}$$

Let $z = \frac{x}{\|s\|_K}$, then,

$$\begin{aligned} (C_m^K(f))^\circ &= \left\{(y,t) \in \mathbb{R}^n \times \mathbb{R}^m; \|s\|_K \langle z,y\rangle + \langle s,t\rangle \leq 1, \forall (z,s), f(z) \leq \frac{1}{\|s\|_K}\right\} \\ &= \left\{(y,t) \in \mathbb{R}^n \times \mathbb{R}^m; \sup_{\{s; \|s\|_K \leq 1/f(z)\}} \|s\|_K \langle z,y\rangle + \langle s,t\rangle \leq 1, \forall z\right\} \\ &= \left\{(y,t) \in \mathbb{R}^n \times \mathbb{R}^m; \frac{\langle z,y\rangle + \|t\|_{K^\circ}}{f(z)} \leq 1, \forall z\right\} \\ &= \left\{(y,t) \in \mathbb{R}^n \times \mathbb{R}^m; \sup_{z\in\mathbb{R}^n} \frac{\langle z,y\rangle + \|t\|_{K^\circ}}{f(z)} \leq 1\right\} \\ &= \overline{\left\{(y,t) \in \mathbb{R}^n \times \mathbb{R}^m \setminus \{0\}; \|t\|_{K^\circ} \mathcal{M}f\left(\frac{y}{\|t\|_{K^\circ}}\right) \leq 1\right\}} \\ &= C_m^{K^\circ}(\mathcal{M}f). \end{aligned}$$

Let us compute the volume of the convex body $C_m^K(f)$.

$$\begin{aligned}\left|C_m^K(f)\right| &= \left|\left\{(x,s)\in\mathbb{R}^n\times\mathbb{R}^m\setminus\{0\};\|s\|_K f\left(\frac{x}{\|s\|_K}\right)\le 1\right\}\right| \\ &= \int_{\mathbb{R}^m}\left|\left\{x\in\mathbb{R}^n;\|s\|_K f\left(\frac{x}{\|s\|_K}\right)\le 1\right\}\right|_n ds \\ &= \int_{\mathbb{R}^m}\|s\|_K^n\left|\left\{z\in\mathbb{R}^n;f(z)\le\frac{1}{\|s\|_K}\right\}\right|_n ds=\int_{\mathbb{R}^m}\varphi(\|v\|_K)dv,\end{aligned}$$

where $\varphi(t)=t^n\left|\left\{z\in\mathbb{R}^n;f(z)\le\frac{1}{t}\right\}\right|_n$. Using Fubini and a change of variables, one has

$$\int_{\mathbb{R}^m}\varphi(\|v\|_K)dv=\int_0^{+\infty}(-\varphi'(t))|\{v\in\mathbb{R}^m;\|v\|_K\le t\}|dt=|K|\int_0^{+\infty}t^m(-\varphi'(t))dt=m|K|\int_0^{+\infty}t^{m-1}\varphi(t)dt.$$

Hence, we get

$$\begin{aligned}|C_m^K(f)| &= m|K|\int_0^{+\infty}t^{m-1}t^n\left|\left\{z\in\mathbb{R}^n;f(z)\le\frac{1}{t}\right\}\right|_n dt \\ &= m|K|\int_0^{+\infty}\frac{1}{u^{m+n+1}}\left|\{z\in\mathbb{R}^n;f(z)\le u\}\right|du \\ &= m|K|\int_{\mathbb{R}^n}\int_{f(z)}^{+\infty}\frac{du}{u^{m+n+1}}dz \\ &= \frac{m}{m+n}|K|\int_{\mathbb{R}^n}\frac{dz}{f(z)^{m+n}}.\end{aligned}$$

Using that $C_m^K(f)^\circ=C_m^{K^\circ}(\mathcal{M}f)$, we get

$$|C_m^K(f)^\circ|=\frac{m}{m+n}|K^\circ|\int_{\mathbb{R}^n}\frac{dz}{(\mathcal{M}f(z))^{m+n}}.$$

Thus, we obtain for any $m>0$

$$\int_{\mathbb{R}^n}\frac{1}{f(x)^{m+n}}dx\int_{\mathbb{R}^n}\frac{1}{(\mathcal{M}f(y))^{m+n}}dy=\left(\frac{m+n}{m}\right)^2\frac{1}{|K||K^\circ|}|C_m^K(f)||C_m^K(f)^\circ|. \tag{2.6}$$

3. Proof of the inequalities in dimension 1 and 2 for $s>0$.

In this section, we start by giving a monotonicity result, which implies Conjecture 1.2 in dimension 1 where $\frac{1}{s}\in\mathbb{N}$. Recall that Conjecture 1.2 in dimension 1 was proved in [FM10, page 17] for any $s>0$. We present here a simpler proof in the case where $\frac{1}{s}\in\mathbb{N}$. Then, we present a proof of Theorem 1.3, which is Mahler's conjecture for s-concave functions in the case $n=2$ and $\frac{1}{s}\in\mathbb{N}$.

Let $m=\frac{1}{s}$ and $g:\mathbb{R}^n\to\mathbb{R}_+$ be an even s-concave function, then there exists an even function $f:\mathbb{R}^n\to\mathbb{R}$, concave on its support, such that $g=f^m$. Thus, in dimension $n=1$ and for $1/s\in\mathbb{N}$, Conjecture 1.2 is a consequence of the following theorem.

Theorem 3.1. *Let $m\in\mathbb{N}$, $a>0$, $f:\mathbb{R}\to\mathbb{R}_+$ be an even concave function on its support $C=[-a,a]$ and*

$$P_m(f)=\int_{\mathbb{R}}f(x)^m dx\int_{\mathbb{R}}(\mathcal{L}f(y))^m dy.$$

Then the sequence $((m+1)^2P_m(f)-4m)_m$ is non-decreasing. Hence, one has $P_m(f)\ge\frac{4}{m+1}$. Moreover, if $P_m(f)=\frac{4}{m+1}$ then either f is constant on its support $[-a,a]$ or there is $c>0$ such that $f(x)=c\left(1-\frac{|x|}{a}\right)_+$.

Proof. First, using Lemma 2.1, we know that $\mathrm{supp}(\mathcal{L}f) = C^\circ = [-\frac{1}{a}, \frac{1}{a}]$. Set

$$I_m = (m+1)\int_0^a f(x)^m dx \quad \text{and} \quad J_m = (m+1)\int_0^{\frac{1}{a}} (\mathcal{L}f(y))^m dy.$$

The monotonicity that we want to prove reduces to $I_m J_m \geq m+1$, for every $m \in \mathbb{N}$. The inequality holds for $m = 0$ (there is equality). Let $m \in \mathbb{N}^*$. Being concave on $[-a, a]$, the function f is left differentiable on $(-a, a]$. By concavity, denoting by $f'(t)$ the left derivative of f at t, we have for all $t \in (-a, a]$ and $x \in [-a, a]$

$$f(x) \leq f(t) + (x-t)f'(t). \tag{3.1}$$

multiplying both sides of the inequality by $mf(t)^{m-1}$ and integrating on $[0, a]$ gives

$$f(x)I_{m-1} = f(x)\int_0^a mf(t)^{m-1}dt \leq \int_0^a mf(t)^m dt + \int_0^a m(x-t)f'(t)f(t)^{m-1}dt.$$

Integration by parts gives that for every $x \in [-a, a]$

$$\int_0^a m(x-t)f'(t)f(t)^{m-1}dt = (x-a)f(a)^m - xf(0)^m + \int_0^a f(t)^m dt.$$

Using that $(x-a)f(a)^m \leq 0$, we get that for all $x \in [-a, a]$,

$$f(x)I_{m-1} \leq I_m - xf(0)^m. \tag{3.2}$$

Hence, one has for all $x \in [-a, a]$

$$\frac{I_{m-1}}{I_m} \leq \frac{1}{f(x)}\left(1 - \frac{xf(0)^m}{I_m}\right).$$

By taking the infinimum over all $x \in [-a, a]$, and denoting $y_0 = \frac{f(0)^m}{I_m}$ we get

$$\frac{I_{m-1}}{I_m} \leq \mathcal{L}f\left(y_0\right).$$

Since $\mathcal{L}f$ is concave, we can apply the result to $\mathcal{L}f$ and, denoting $x_0 = \frac{(\mathcal{L}f(0))^m}{J_m}$, we get

$$\frac{J_{m-1}}{J_m} \leq f\left(x_0\right).$$

On the other hand, the definition of $\mathcal{L}f$ implies that for all $x \in \mathrm{supp} f$, $y \in \mathrm{supp}\mathcal{L}f$

$$f(x)\mathcal{L}f(y) \leq (1-xy)_+.$$

Using (3.2) for $x = 0$ gives that $I_m \geq f(0)I_{m-1}$, thus $I_m \geq f(0)^m a$. In the same way, one has $J_m \geq \mathcal{L}f(0)^m a^{-1}$. Multiplying these inequalities and using that f is even and concave, we get $\mathcal{L}f(0) = \max(f)^{-1} = f(0)^{-1}$, we conclude that $I_m J_m \geq 1$, which implies that $x_0 y_0 \leq 1$. Hence, one has

$$\frac{I_{m-1}J_{m-1}}{I_m J_m} \leq f\left(x_0\right)\mathcal{L}f\left(y_0\right) \leq 1 - x_0 y_0 = 1 - \frac{1}{I_m J_m}. \tag{3.3}$$

Finally we get $I_{m-1}J_{m-1} \leq I_m J_m - 1$. Hence the sequence $(I_m J_m - m)_m$ is non-decreasing. Since $I_m J_m = \frac{(m+1)^2}{4}P_m(f)$, we deduce that the sequence $u_m = (m+1)^2 P_m(f) - 4m$ is non-decreasing. It follows that $u_m \geq u_0 = P_0(f) = 4$, hence $P_m(f) \geq \frac{4}{m+1}$, for any integer m.

Assume now that there is equality, for some integer m: $P_m(f) = \frac{4}{m+1}$. Then, there is equality in (3.3). It follows that there is equality in (3.2) for $x = x_0$ and also $(x_0 - a)f(a)^m = 0$. Moreover, there is also equality in the corresponding inequality for $\mathcal{L}f$ instead of f and y_0 instead of x_0, thus $(y_0 - \frac{1}{a})\mathcal{L}f(1/a)^m = 0$. In addition, there is equality in (3.1) for $x = x_0$ and for almost any $t \in [-a, a]$:

$$f(t) = f(x_0) + (t - x_0)f'(t).$$

On the other hand, one has $f(t) = f(x_0) + \int_{x_0}^t f'(u)du$. Since the left derivative f' is non-increasing, we deduce that f' is constant on $[0, x_0)$ and on $(x_0, a]$, which means that f is affine on on $[0, x_0)$ and on $(x_0, a]$. In the same way, $\mathcal{L}f$ is affine on $[0, y_0)$ and on $(y_0, 1/a]$.

If $x_0 = a$, then f is affine on $[0, a]$ and $J_m = \mathcal{L}f(0)^m/a$. But, by concavity, one has for every $y \in [0, 1/a]$

$$\mathcal{L}f(y) \geq \mathcal{L}f(0)(1 - ay).$$

Raising this inequality to the power m and integrating, one gets that $J_m \geq \mathcal{L}f(0)^m/a$. Since there is equality we deduce that $\mathcal{L}f(y) = \mathcal{L}f(0)(1 - ay)$, for every $y \in [0, 1/a]$. Thus $f = f(0)\mathbb{1}_{[-a,a]}$.

If $x_0 \neq a$, then $f(a) = 0$. Thus there exists $0 \leq b \leq 1/a$ and $c > 0$ such that, for $t \geq 0$,

$$f(t) = c(1 - bt)\mathbb{1}_{[0,x_0]} + c(1 - bx_0)\frac{a - x}{a - x_0}\mathbb{1}_{(x_0,a]}.$$

This implies that, for any $y \geq 0$,

$$\mathcal{L}f(y) = \frac{1}{c}\left(\mathbb{1}_{[0,b]}(y) + \frac{1 - x_0 y}{1 - bx_0}\mathbb{1}_{(b,\frac{1}{a}]}(y)\right).$$

Since $x_0 \neq a$, one has $\mathcal{L}f(1/a) \neq 0$, hence $y_0 = 1/a$. This gives $I_m = af(0)^m$ and this implies as before that $f(x) = f(0)(1 - \frac{x}{a})$, for all $x \in [0, a]$.

□

Let us prove the 2-dimensional case of our inequality for m being an integer.

Theorem 3.2. *Let $m \in \mathbb{N}$ and $f : \mathbb{R}^2 \to \mathbb{R}_+$ be an even function, concave on its support, such that $0 < \int f < +\infty$. Then,*

$$P_m(f) \geq \frac{16}{(m + 1)(m + 2)}.$$

Moreover, there is equality for $f = \mathbb{1}_{[-1,1]^2}$.

As noticed before, this theorem may be reformulated in the following form: for any integer $m > 0$ and any even function $g : \mathbb{R}^2 \to \mathbb{R}_+$ that is $\frac{1}{m}$-concave, integrable with $\int_{\mathbb{R}^2} g > 0$ one has

$$\int_{\mathbb{R}^2} g \int_{\mathbb{R}^2} \mathcal{L}_{\frac{1}{m}} g \geq \frac{16m^2}{(m + 1)(m + 2)}. \tag{3.4}$$

The following corollary proves that letting m go to infinity we get a new proof of Mahler conjecture for even log-concave functions, which was our previous main result in [FN].

Corollary 3.3. *Let $f : \mathbb{R}^2 \to \mathbb{R}$ be an even log-concave function such that $0 < \int f < +\infty$, then*

$$\int_{\mathbb{R}^2} f(x)dx \int_{\mathbb{R}^2} f^\circ(y)dy \geq 16, \tag{3.5}$$

where $f^\circ(y) = \inf_{x \in \mathbb{R}^n} \frac{e^{-\langle x,y\rangle}}{f(x)}$ is the polar function of f.

Proof. Let f be an even log-concave function and consider the function f_m defined as follows

$$f_m(x) = \left(1 + \frac{\log f(x)}{m}\right)_+^m.$$

First, notice that the log-concavity of f implies the $\frac{1}{m}$-concavity of f_m. Since for all x, $e^{-x} \geq 1 - x$, we get

$$\left(1 - \frac{1}{m}\langle x, y\rangle\right)^m \leq e^{-\langle x,y\rangle}.$$

Thus, $\mathcal{L}_{\frac{1}{m}} f \leq \mathcal{L}_0 f = f^\circ$. In addition, it is easy to see that $f_m \leq f$ for any $m > 0$, and since a log-concave function is continuous on its support one has $\lim_{m\to\infty} f_m = f$ locally uniformly on

$\mathbb{R}^2$. Hence, we can conclude that $\lim_{m\to\infty}\int_{\mathbb{R}^2} f_m = \int_{\mathbb{R}^2} f$. The fact that $\mathcal{L}_{\frac{1}{m}} f \leq f^\circ$ implies that $\mathcal{L}_{\frac{1}{m}} f_m \leq f_m^\circ$. Moreover, one has

$$\frac{16m^2}{(m+1)(m+2)} \leq \int_{\mathbb{R}^2} f_m \int_{\mathbb{R}^2} \mathcal{L}_{\frac{1}{m}} f_m \leq \int_{\mathbb{R}^2} f_m \int_{\mathbb{R}^2} f_m^\circ.$$

On the other hand, since the functions f_m are log-concave and $\lim_{m\to\infty} f_m = f$, then using lemma 3.2 in [AKM], we know that $\lim_{m\to\infty} f_m^\circ = f^\circ$ locally uniformly on the interior of the support of f°. Applying the same lemma again on the log-concave function f_m°, we get that $\lim_{m\to\infty}\int_{\mathbb{R}^2} f_m^\circ = \int_{\mathbb{R}^2} f^\circ$. Thus, we conclude that

$$\int_{\mathbb{R}^2} f \int_{\mathbb{R}^2} f^\circ = \lim_{m\to\infty} \int_{\mathbb{R}^2} f_m \int_{\mathbb{R}^2} f_m^\circ \geq 16.$$

□

Remark 3.4. Notice that we don't get the equality case in this way. Recall that, in [FN], it was proved that there is equality in (3.5) if and only if there exists two Hanner polytopes $K_1 \subset F_1$ and $K_2 \subset F_2$, where F_1 and F_2 are two supplementary subspaces in $\mathbb{R}^2$, with $0 \leq \dim(F_i) \leq 2$, such that for all $(x_1, x_2) \in F_1 \times F_2$

$$f(x_1, x_2) = e^{-\|x_1\|_{K_1}} \mathbb{1}_{K_2}(x_2).$$

It would be natural to conjecture that a concave function f on its support satisfies the equality case in (3.4) if and only if

$$f(x_1, x_2) = (1 - \|x_1\|_{K_1})_+ \mathbb{1}_{K_2}(x_2),$$

with K_1, K_2, F_1, F_2 as before. It is easy to see that such functions satisfy the equality case. Nevertheless, clearly, the function $f(x_1, x_2) = (1 - |x_1|)_+ \mathbb{1}_{B_1^2}(x_1, x_2)$ satisfies also the equality case but cannot be written in this form.

For the proof of theorem 3.2, let us start by proving that every even concave function $f : \mathbb{R}^2 \to \mathbb{R}_+$ can be m-equipartitioned by the canonical basis of $\mathbb{R}^2$.

Lemma 3.5. *Let $m > 0$ and $f : \mathbb{R}^2 \to \mathbb{R}_+$ be an even concave function such that $0 < \int_{\mathbb{R}^2} f < +\infty$. Then there exists a linear invertible map $T : \mathbb{R}^2 \to \mathbb{R}^2$ such that the function $f \circ T$ is m-equipartitioned.*

Proof. The proof follows the proof of Lemma 4.2 in our previous work [FN] but we reproduce it for completeness. For any $u \in S^1$, let $C(u) \subset S^1$ be the open half-circle delimited by u and $-u$ containing the vectors v which are after u with respect to the counterclockwise orientation of S^1. For $v \in C(u)$, let $C(u, v) = \mathbb{R}_+ u + \mathbb{R}_+ v$ be the cone generated by u and v and define

$$f_u(v) = \int_{C(u,v)} f(x)^m dx.$$

The map f_u is continuous and increasing on $C(u)$, $f_u(u) = 0$ and, since f is even, $f_u(v) \longrightarrow \frac{1}{2}\int_{\mathbb{R}^2} f^m(x)dx$ when $v \longrightarrow -u$. Thus, there exists a unique $v(u) \in C(u)$ such that

$$f_u(v(u)) = \int_{C(u,v(u))} f^m(x)dx = \frac{1}{4}\int_{\mathbb{R}^2} f^m(x)dx.$$

Notice that $v : S^1 \to S^1$ is continuous and, since f is even, one has

$$\int_{C(v(u),-u)} f^m(x)dx = \frac{1}{4}\int_{\mathbb{R}^2} f^m(x)dx,$$

thus $v(v(u)) = -u$ for any $u \in S^1$. For $u \in S^1$, let

$$g(u) = \int_{C(u,v(u))} f^{m-1}(x)dx - \frac{1}{4}\int_{\mathbb{R}^2} f^{m-1}(x)dx.$$

Then, g is continuous on S^1 and, since f is even,

$$\begin{aligned} g(v(u)) &= \int_{C(v(u),-u)} f^{m-1}(x)dx - \frac{1}{4}\int_{\mathbb{R}^2} f^{m-1}(x)dx \\ &= \frac{1}{2}\int_{\mathbb{R}^2} f^{m-1}(x)dx - \int_{C(u,v(u))} f^{m-1}(x)dx - \frac{1}{4}\int_{\mathbb{R}^2} f^{m-1}(x)dx \\ &= \frac{1}{4}\int_{\mathbb{R}^2} f^{m-1}(x)dx - \int_{C(u,v(u))} f^{m-1}(x)dx \\ &= -g(u). \end{aligned}$$

Hence, by the intermediate value theorem, there exists $u \in S^1$ such that $g(u) = 0$, thus

$$\int_{C(u,v(u))} f^{m-1} = \frac{1}{4}\int_{\mathbb{R}^2} f^{m-1} \quad \text{and} \quad \int_{C(u,v(u))} f^{m} = \frac{1}{4}\int_{\mathbb{R}^2} f^{m}.$$

□

Let us proceed to the proof of our main theorem.

Proof of Theorem 3.2. Let $f : \mathbb{R}^2 \to \mathbb{R}_+$ be an even concave function such that $0 < \int_{\mathbb{R}^2} f < +\infty$. Let $L = B_\infty^m$. For simplicity, the convex body $K_{B_\infty^m,m}(f)$ introduced in section 2 will be denoted in this case by $K_{\infty,m}(f)$ and its dual $K_{B_1^m,m}(\mathcal{L}f)$ will be denoted by $K_{1,m}(\mathcal{L}f)$. Thus, we know that

$$P(K_{\infty,m}(f)) = P(B_\infty^m)P_m(f) = \frac{4^m}{m!}P_m(f).$$

The proof of the theorem concludes if $K_{\infty,m}$ verifies Mahler's conjecture in dimension $m+2$, i.e. $P(K_{\infty,m}) \geq \frac{4^{m+2}}{(m+2)!}$. For that, it is enough to prove that the convex body $K_{\infty,m}$ verifies the hypotheses of Proposition 1.11. We proceed by induction on $m \in \mathbb{N}$. Set, for any positive integer k and for any $\varepsilon \in \{-1,1\}^k$, $\mathbb{R}^k_\varepsilon = \{x \in \mathbb{R}^k; \varepsilon_i x_i \geq 0, \forall i \in \{1,...,k\}\}$.

- Let us start by proving that $K_{\infty,m}(f)$ can be partitioned into 2^{m+2} pieces of the same volume. Since f is m-equipartitioned, we get

$$\begin{aligned} \left|K_{\infty,m}(f) \cap \mathbb{R}^{m+2}_+\right| &= \left|\{(x,t) \in \mathbb{R}^2_+ \times \mathbb{R}^m_+; \|t\|_\infty \leq f(x)\}\right| = \int_{\mathbb{R}^2_+} f(x)^m dx |B_\infty^m \cap \mathbb{R}^m_+| \\ &= \int_{\mathbb{R}^2_+} f(x)^m dx = \frac{1}{2^2|B_\infty^m|}\int_{\mathbb{R}^2} f(x)^m |B_\infty^m| dx = \frac{1}{2^{m+2}}|K_{\infty,m}(f)|. \end{aligned}$$

  Using the fact that $\int_{\mathbb{R}^2_+} f(x)^m dx = \int_{\mathbb{R}_+\times\mathbb{R}_-} f(x)^m dx$, we get,

$$|K_{\infty,m}(f) \cap \mathbb{R}^{m+2}_\varepsilon| = \frac{1}{2^{m+2}}|K_{\infty,m}(f)| \quad \forall \varepsilon \in \{-1,1\}^{m+2}.$$

- Our next step is to prove that $K_{\infty,m}(f) \cap e_i^\perp$ verifies Mahler's conjecture for all $i \in \{1,...,m+2\}$.
  For $i = 1$ or $i = 2$, the same argument holds so let us assume that $i = 1$. One has

$$K_{\infty,m}(f) \cap e_1^\perp = \{(0,x_2,t) \in \mathbb{R}^2 \times \mathbb{R}^m; \|t\|_\infty \leq f(0,x_2)\}.$$

  Thus,

$$|K_{\infty,m}(f) \cap e_1^\perp| = 2^m \int_{\mathbb{R}} f(0,x_2)^m dx_2.$$

  Since, the dual of a section is the projection of the dual, one has

$$\left(K_{\infty,m}(f) \cap e_1^\perp\right)^\circ = \left(K_{\infty,m}\left(f_{|e_1^\perp}\right)\right)^\circ = K_{1,m}\left(\mathcal{L}\left(f_{|e_1^\perp}\right)\right).$$

Thus

$$\left|\left(K_{\infty,m}(f)\cap e_1^{\perp}\right)^{\circ}\right| = \frac{2^m}{m!}\int_{\mathbb{R}}\left(\mathcal{L}\left(f_{|e_1^{\perp}}\right)(y)\right)^m dy.$$

Our Theorem 3.1 in dimension 1 implies that

$$\int_{\mathbb{R}}(f(0,x_2))^m dx_2 \int_{\mathbb{R}}(\mathcal{L}(f_{|e_1^{\perp}})(y))^m dy \geq \frac{4}{m+1}.$$

Hence,

$$P(K_{\infty,m}(f)\cap e_1^{\perp}) = \frac{4^m}{m!}\int_{\mathbb{R}}(f(0,x_2))^m dx_2 \int_{\mathbb{R}}(\mathcal{L}(f_{|e_1^{\perp}})(y))^m dy \geq \frac{4^{m+1}}{(m+1)!}.$$

which implies that $K_{\infty,m}(f)\cap e_1^{\perp}$ verifies Mahler's conjecture in $\mathbb{R}^{m+1}$.
Now, for $i\geq 3$, again, by symmetry, we may assume that $i=3$. One has

$$K_{\infty,m}(f)\cap e_3^{\perp} = \{(x,t)\in\mathbb{R}^2\times(\mathbb{R}^m\cap e_3^{\perp}); \|t\|_{\infty}\leq f(x)\} = K_{\infty,m-1}(f).$$

Hence,

$$\begin{aligned} P(K_{\infty,m}(f)\cap e_3^{\perp}) &= P(K_{\infty,m-1}(f)) = \frac{4^{m-1}}{(m-1)!}\int_{\mathbb{R}^2} f(x)^{m-1}dx \int_{\mathbb{R}^2}(\mathcal{L}f(y))^{m-1}dy \\ &\geq \frac{4^{m-1}}{(m-1)!}\frac{16}{m(m+1)} = \frac{4^{m+1}}{(m+1)!}. \end{aligned}$$

where the last inequality occurs by the induction hypothesis.

- Finally, let us show that $K_{\infty,m}(f)\cap e_i^{\perp}$, for any $i\in\{1,...,m+1\}$, can be partitioned into 2^{m+1} pieces of the same $(m+1)$-volume. For $i=1$ (and the same follows for $i=2$), since f is even, one has

$$\begin{aligned} \left|K_{\infty,m}(f)\cap e_1^{\perp}\cap\mathbb{R}_+^{m+1}\right|_{m+1} &= \left|\{(0,x_2,t)\in\mathbb{R}_+^2\times\mathbb{R}_+^m; \|t\|_{\infty}\leq f(0,x_2)\}\right|_{m+1} \\ &= \int_0^{+\infty}\left|f(0,x_2)B_{\infty}^m\cap\mathbb{R}_+^m\right|_m dx_2 \\ &= \int_0^{\infty} f(0,x_2)^m dx_2 \\ &= \frac{1}{2^{m+1}}\left|K_{\infty,m}(f)\cap e_1^{\perp}\right|_{m+1}. \end{aligned}$$

For $i=3$ (and the same follows for $i>3$),

$$K_{\infty,m}(f)\cap e_3^{\perp}\cap\mathbb{R}_+^{m+1} = \{(x,t)\in\mathbb{R}_+^2\times(\mathbb{R}_+^m\cap e_1^{\perp}); \|t\|_{\infty}\leq f(x)\} = K_{\infty,m-1}(f)\cap\mathbb{R}_+^{m+1}.$$

Thus,

$$\left|K_{\infty,m}(f)\cap e_3^{\perp}\cap\mathbb{R}_+^{m+1}\right|_{m+1} = \int_{\mathbb{R}_+^2} f(x)^{m-1}dx\times|B_{\infty}^{m-1}\cap\mathbb{R}_+^{m-1}|_{m-1} = \int_{\mathbb{R}_+^2} f(x)^{m-1}dx,$$

and the result follows from the m-equipartition condition.

□

Remark 3.6. Notice that if one replaces in the above argument the cube B_{∞}^m by any unconditional convex body L in $\mathbb{R}^m$, the same proof shows that $K_{L,m}(f)$ satisfies Mahler's conjecture in dimension $m+2$.

4. Proof of the inequalities in dimension 1 and 2 for $-\frac{1}{n} < s < 0$.

Let $-\frac{1}{n} < s = -\frac{1}{m} < 0$ and $g : \mathbb{R}^n \to \mathbb{R}_+$ be an even s-concave function. Then, the function $f = g^s : \mathbb{R}^n \to (0, +\infty]$ is even and convex. In this section, we give a sharp lower bound of the operator Q_m applied to even convex functions f in dimensions 1 and 2. First, we start by proving the following theorem, which is the analogue version of Theorem 1.6 for convex functions rather than s-concave ones, as seen in equation (2.4).

Theorem 4.1. *Let $f : \mathbb{R}^2 \to (0, +\infty)$ be an even convex function such that $f \in \mathcal{F}$. Then, for any integer $m > 2$, one has*

$$\int_{\mathbb{R}^2} \frac{1}{(f(x))^m} dx \int_{\mathbb{R}^2} \frac{1}{(\mathcal{M}f(y))^m} dy \geq \frac{16m}{(m-1)(m-2)^2}.$$

Moreover, there is equality for $f(x) = \max(1, \|x\|_\infty)$.

Proof. Let us recall equation (2.6) and apply it in dimension n, which gives that, for all $m > n$ and for every symmetric convex body $K \in \mathbb{R}^{m-n}$, one has

$$\int_{\mathbb{R}^n} \frac{1}{f(x)^m} dx \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}f(y))^m} dy = \left(\frac{m}{m-n}\right)^2 \frac{1}{|K||K^\circ|} |C^K_{m-n}(f)||C^K_{m-n}(f)^\circ|,$$

where $C^K_{m-n}(f) = \left\{(x, s) \in \mathbb{R}^n \times \mathbb{R}^{m-n}; \|s\|_K f\left(\frac{x}{\|s\|_K}\right) \leq 1\right\}$. Applying it for $K = B^{m-n}_\infty$, we simplify the notation of the associated convex body to

$$C^\infty_{m-n}(f) = \left\{(x, s) \in \mathbb{R}^n \times \mathbb{R}^{m-n}; \|s\|_\infty f\left(\frac{x}{\|s\|_\infty}\right) \leq 1\right\}.$$

And its dual is denoted by $C^1_{m-n}(\mathcal{M}f)$. In addition, if $C^\infty_{m-n}(f)$ verifies Mahler's inequality in dimension m, we get

$$\begin{aligned}
\int_{\mathbb{R}^n} \frac{1}{f(x)^m} dx \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}f(y))^m} dy &= \left(\frac{m}{m-n}\right)^2 \frac{1}{|B^{m-n}_\infty||B^{m-n}_1|} |C^\infty_{m-n}(f)||C^\infty_{m-n}(f)^\circ| \\
&\geq \left(\frac{m}{m-n}\right)^2 \frac{(m-n)!}{4^{m-n}} \frac{4^m}{m!} \\
&= \frac{m}{m-n} \times \frac{4^n}{(m-n)\cdots(m-1)}.
\end{aligned}$$

Thus, we get the desired inequality (2.4) in dimension n. Since the m-equipartition condition, which is the key of our proof, is only verified in dimension 2, it remains to prove that $C^\infty_{m-2}(f)$ verifies the hypothesis of Proposition 1.11 in dimension m for any integer $m \geq 3$.

We prove it by induction on m. For $m = 3$, $C^\infty_{m-2}(f) = C^\infty_1(f)$ is a symmetric convex body in $\mathbb{R}^3$ hence it satisfies Mahler's conjecture by [IS].

Let $m \geq 4$ and assume that Mahler's conjecture is satisfied for $C^\infty_{m-3}(f)$ and let us prove it for $C^\infty_{m-2}(f)$. For that, recall that $\mathbb{R}^2_\varepsilon = \{x \in \mathbb{R}^2; \varepsilon_i x_i \geq 0, \forall i \in \{1, 2\}\}$, $\forall \varepsilon \in \{-1, 1\}^2$. We can assume that the even convex function $f : \mathbb{R}^2 \to \mathbb{R}_+$ is m-equipartitioned. In the sense that, $\forall \varepsilon \in \{-1, 1\}^2$, one has

$$\int_{\mathbb{R}^2_\varepsilon} \frac{dx}{f(x)^m} = \frac{1}{4} \int_{\mathbb{R}^2} \frac{dx}{f(x)^m} \quad \text{and} \quad \int_{\mathbb{R}^2_\varepsilon} \frac{dx}{f(x)^{m-1}} = \frac{1}{4} \int_{\mathbb{R}^2} \frac{dx}{f(x)^{m-1}}.$$

We use proposition 1.11 by proving that the convex body $C^\infty_{m-2}(f)$ verifies its conditions to deduce the lower bound of Mahler's volume of $C^\infty_{m-2}(f)$.

- <u>$C^\infty_{m-2}(f)$ can be partitioned into 2^m pieces of the same volume.</u>

$$C^\infty_{m-2}(f) \cap \mathbb{R}^m_+ = \left\{(x, s) \in \mathbb{R}^2_+ \times \mathbb{R}^{m-2}_+; \|s\|_\infty f\left(\frac{x}{\|s\|_\infty}\right) \leq 1\right\}.$$

Then,

$$\left|C^{\infty}_{m-2}(f)\cap\mathbb{R}^m_+\right| = \frac{m-2}{m}\left|B^{m-2}_{\infty}\cap\mathbb{R}^2_+\right|\int_{\mathbb{R}^2_+}\frac{dx}{f(x)^m} = \frac{m-2}{4m}\int_{\mathbb{R}^2}\frac{dx}{f(x)^m}.$$

Thus,

$$|C^{\infty}_{m-2}(f)| = \frac{m-2}{m}|B^{m-2}_{\infty}|\int_{\mathbb{R}^2}\frac{dx}{f(x)^m} = \frac{m-2}{m}2^{m-2}\int_{\mathbb{R}^2}\frac{dx}{f(x)^m} = 2^m|C^{\infty}_{m-2}(f)\cap\mathbb{R}^m_+|.$$

Similarly, since f is even and m-equipartitioned, we can easily prove that for all $\varepsilon\in\{-1,1\}^m$

$$|C^{\infty}_{m-2}(f)\cap\mathbb{R}^m_{\varepsilon}| = \frac{1}{2^m}|C^{\infty}_{m-2}(f)|.$$

- <u>$C^{\infty}_{m-2}(f)\cap e_i^{\perp}$ verifies Mahler in dimension $m-1$, for $1\le i\le m$.</u>

  For $i=1$ (the same holds for $i=2$),

$$C^{\infty}_{m-2}(f)\cap e_1^{\perp} = \left\{(0,x_2,s)\in\mathbb{R}^2\times\mathbb{R}^{m-2};\|s\|_{\infty}f\left(\frac{(0,x_2)}{\|s\|_{\infty}}\right)\le 1\right\} = \{0\}\times C^{\infty}_{m-2}\left(f_{|e_1^{\perp}}\right).$$

One has,

$$\left(C^{\infty}_{m-2}(f)\cap e_1^{\perp}\right)^{\circ} = \left(\{0\}\times C^{\infty}_{m-2}\left(f_{|e_1^{\perp}}\right)\right)^{\circ} = \{0\}\times\left(C^{\infty}_{m-2}\left(f_{|e_1^{\perp}}\right)\right)^{\circ} = \{0\}\times C^{1}_{m-2}\left(\mathcal{M}\left(f_{|e_1^{\perp}}\right)\right).$$

Thus, we get the following volumes

$$\left|C^{\infty}_{m-2}(f)\cap e_1^{\perp}\right| = \left|C^{\infty}_{m-2}\left(f_{e_1^{\perp}}\right)\right| = \frac{m-2}{m-1}2^{m-2}\int_{\mathbb{R}}\frac{dx_2}{f_{|e_1^{\perp}}(x_2)^{m-1}}.$$

And

$$\left|\left(C^{\infty}_{m-2}(f)\cap e_1^{\perp}\right)^{\circ}\right| = \left|C^{1}_{m-2}\left(\mathcal{M}\left(f_{|e_1^{\perp}}\right)\right)\right| = \frac{(m-2)2^{m-2}}{(m-1)(m-2)!}\int_{\mathbb{R}}\frac{dx_2}{\left(\mathcal{M}f_{|e_1^{\perp}}(x_2)\right)^{m-1}}.$$

Applying the result in dimension 1, proved in [FGSZ], to $f_{|e_1^{\perp}}$ we get

$$\begin{aligned}\left|C^{\infty}_{m-2}(f)\cap e_1^{\perp}\right|\left|\left(C^{\infty}_{m-2}(f)\cap e_1^{\perp}\right)^{\circ}\right| &= \frac{(m-2)^2 4^{m-2}}{(m-1)^2(m-2)!}\int_{\mathbb{R}}\frac{dx_2}{f_{|e_1^{\perp}}(x_2)^{m-1}}\int_{\mathbb{R}}\frac{dx_2}{\left(\mathcal{M}f_{|e_1^{\perp}}(x_2)\right)^{m-1}}\\ &\ge \frac{(m-2)^2 4^{m-2}}{(m-1)^2(m-2)!}\frac{m-1}{m-2}\frac{4}{m-2)} = \frac{4^{m-1}}{(m-1)!}.\end{aligned}$$

For $i=3$ and the case $i>3$ follows similarly:

$$C^{\infty}_{m-2}(f)\cap e_3^{\perp} = \left\{(x,0,s)\in\mathbb{R}^2\times\{0\}\times\mathbb{R}^{m-3};\|(0,s)\|_{\infty}f\left(\frac{x}{\|(0,s)\|_{\infty}}\right)\le 1\right\} = C^{\infty}_{m-3}(f).$$

Thus,

$$\left|C^{\infty}_{m-2}(f)\cap e_3^{\perp}\right| = \left|C^{\infty}_{m-3}(f)\right| = \frac{m-3}{m-1}2^{m-3}\int_{\mathbb{R}^2}\frac{dx}{f(x)^{m-1}}.$$

And

$$\left|\left(C^{\infty}_{m-2}(f)\cap e_3^{\perp}\right)^{\circ}\right| = \left|\left(C^{\infty}_{m-3}(f)\right)^{\circ}\right| = \left|C^{1}_{m-3}(\mathcal{M}f)\right| = \frac{m-3}{m-1}\frac{2^{m-3}}{(m-3)!}\int_{\mathbb{R}^2}\frac{dy}{(\mathcal{M}f(y))^{m-1}}.$$

Then, using the hypothesis of induction on m, we obtain,

$$\begin{aligned}\left|C^\infty_{m-2}(f)\cap e_3^\perp\right|\left|\left(C^\infty_{m-2}(f)\cap e_3^\perp\right)^\circ\right| &= \frac{(m-3)^2}{(m-1)^2}\frac{4^{m-3}}{(m-3)!}\int_{\mathbb{R}^2}\frac{dx}{f(x)^{m-1}}\int_{\mathbb{R}^2}\frac{dy}{(\mathcal{M}f(y))^{m-1}}\\ &\geq \frac{4^{m-1}}{(m-1)!}.\end{aligned}$$

which concludes the second part.

- <u>$C^\infty_{m-2}(f)\cap e_i^\perp$ can be partitioned into 2^{m-1} pieces of the same volume, for $1\leq i\leq m$.</u>

For $i=1$ and $i=2$, one has

$$C^\infty_{m-2}(f)\cap e_1^\perp=\{0\}\times C^\infty_{m-2}\left(f_{|e_1^\perp}\right)\quad\text{and}\quad C^\infty_{m-2}(f)\cap e_2^\perp=C^\infty_{m-2}\left(f_{|e_2^\perp}\right)\times\{0\}.$$

These convex bodies are unconditional, thus we get the partition.

For $i=3$, by the m-equipartition condition of f, we obtain

$$\begin{aligned}\left|C^\infty_{m-2}(f)\cap e_3^\perp\cap\mathbb{R}^{m-1}_+\right| &= \left|C^\infty_{m-3}(f)\cap\mathbb{R}^{m-1}_+\right|=\frac{m-3}{m-1}\left|B^{m-3}_\infty\cap\mathbb{R}^{m-1}_+\right|\int_{\mathbb{R}^2_+}\frac{dx}{f(x)^{m-1}}\\ &= \frac{1}{4}\frac{m-3}{m-1}\int_{\mathbb{R}^2}\frac{dx}{f(x)^{m-1}}=\frac{1}{2^{m-1}}\left|C^\infty_{m-2}(f)\cap e_3^\perp\right|.\end{aligned}$$

Using the same method, one may prove the partition of the body $C^\infty_{m-2}(f)\cap e_i^\perp$ for $i>3$. Therefore $C^\infty_{m-2}(f)$ satisfies Mahler's conjecture, which concludes the proof of the inequality of Theorem 4.1. Moreover, it is easy to check that there is equality for $f(x)=\max(1,\|x\|_\infty)$, since, in this case $C^\infty_{m-2}(f)=B^m_\infty$. □

Our next goal in this section is to prove the following theorem in dimension 1, which establishes Mahler's conjecture in dimension 1 for s-concave functions for any $-1<s<0$.

Theorem 4.2. *Let $f:\mathbb{R}\to(0,+\infty]$ be an even convex function such that $\lim_{t\to+\infty}f(tx)=+\infty$, for any $x\neq 0$. Then, for any $m>1$, one has*

$$Q_m(f)=\int_{\mathbb{R}}\frac{dx}{f(x)^m}\int_{\mathbb{R}}\frac{dy}{(\mathcal{M}f(y))^m}\geq Q_m\left(1+I_{[-1,1]}\right)=\frac{4}{m-1},$$

with equality if and only if $f=c+I_{[-\alpha,\alpha]}$, for some $c,\alpha>0$.

We first show that Theorem 4.2 follows from the next theorem, which implies Theorem 1.8, if one applies it to $f=g^s$, for any g, s-concave, for $-1<s<0$ and $m=-1/s$.

Theorem 4.3. *Let $f:\mathbb{R}\to(0,+\infty]$ be an even convex function such that $\lim_{t\to+\infty}f(tx)=+\infty$, for any $x\neq 0$. Then for any $m>1$,*

$$\int_{\mathbb{R}}\frac{1}{f(x)^m}dx\geq\frac{m-1}{m}\int_{\mathbb{R}}\frac{1}{(\mathcal{M}\mathcal{M}f(x))^m}dx,\tag{4.1}$$

with equality if and only if $f=c+I_{[-\alpha,\alpha]}$, for some $c,\alpha>0$.

Multiplying equation (4.1) by $\int(\mathcal{M}f)^{-m}$ we deduce that for any even convex f tending to $+\infty$ at infinity, one has

$$Q_m(f)\geq\frac{m-1}{m}Q_m(\mathcal{M}f),\tag{4.2}$$

with equality if and only if $f=c+I_{[-\alpha,\alpha]}$, for some $c,\alpha>0$. Since $\mathcal{M}f\in\mathcal{F}$, using [FGSZ] we get

$$Q_m(\mathcal{M}f)\geq Q_m\left(1+|x|\right)=\frac{4m}{(m-1)^2}.\tag{4.3}$$

Using equations (4.2) and (4.3), we conclude the proof of Theorem 4.2.

Proof of Theorem 4.3. Let us define a variant of the convex body C_m^K introduced in section 2, which is more suitable for $m = 1/|s| \notin \mathbb{N}$, this variant was considered in [IW] for $s > 0$:

$$C_1(f) = C_1^{[-1,1]}(f) = \{(x,t) \in \mathbb{R} \times \mathbb{R}; \pi_1(f)(x,t) \leq 1\} = \left\{(x,t) \in \mathbb{R} \times \mathbb{R}; |t| f\left(\frac{x}{|t|}\right) \leq 1\right\}.$$

The same computations as before show easily that $C_1(f)^\circ = C_1(\mathcal{M}f)$. Moreover, for any $m > 1$, one has

$$\int_{\mathbb{R}} \frac{1}{f(x)^m} dx = \int_{\mathbb{R}} \int_{f(x)}^{+\infty} \frac{m}{t^{m+1}} dt dx = \int_0^{+\infty} \frac{m}{t^{m+1}} |\{x \in \mathbb{R}; f(x) \leq t\}| dt.$$

Using the change of variables $t = 1/u$, we get

$$\int_{\mathbb{R}} \frac{1}{f(x)^m} dx = \frac{m}{2} \int_{C_1(f)} |u|^{m-2} du dx = \mu_m(C_1(f)),$$

where we denote by μ_m the measure with density $d\mu_m(x,t) = \frac{m}{2}|t|^{m-2} dx dt$ on $\mathbb{R}^2$. Applied to $\mathcal{M}\mathcal{M}f$ instead of f, we get

$$\int_{\mathbb{R}} \frac{dz}{(\mathcal{M}\mathcal{M}f(z))^m} = \frac{m}{2} \int_{C_1(\mathcal{M}\mathcal{M}f)} |s|^{m-2} ds dx = \mu_m(C_1(\mathcal{M}\mathcal{M}f)).$$

Since $C_1(\mathcal{M}\mathcal{M}f) = C_1(\mathcal{M}f)^\circ = C_1(f)^{\circ\circ} = \mathrm{Conv}(C_1(f))$, we get

$$\int_{\mathbb{R}} \frac{dz}{(\mathcal{M}\mathcal{M}f(z))^m} = \frac{m}{2} \int_{\mathrm{Conv}(C_1(f))} |s|^{m-2} ds dx = \mu_m(\mathrm{Conv}(C_1(f))).$$

The inequality (4.1) that we want to prove is thus equivalent to

$$\mu_m(C_1(f)) \geq \frac{m-1}{m} \mu_m\left(\mathrm{Conv}(C_1(f))\right). \tag{4.4}$$

It was proved in [FGSZ, Theorem 2.16] that $C_1(f)$ can be written as $C_1(f) = C_1(f)_+ \cup C_1(f)_-$, where

$$C_1(f)_+ = C_1(f) \cap \{s \geq 0\} = \overline{\left\{(x,s) \in \mathbb{R} \times (0,+\infty); s f\left(\frac{x}{s}\right) \leq 1\right\}}$$

is a convex body containing 0 on its boundary and $C_1(f)_-$ is its symmetric image with respect to the hyperplane $\{s = 0\}$. Thus, our problem can be formulated as follows:
for every convex body $K \subset \{(x,s) \in \mathbb{R}^2; s \geq 0\}$ such that $0 \in \partial K$ and K symmetric with respect to $\mathbb{R}e_2$ and if we denote $C = K \cup \sigma(K)$, where $\sigma = \sigma_{e_2{}^\perp}$ then

$$\mu_m(C) \geq \frac{m-1}{m} \mu_m(\mathrm{Conv}(C)). \tag{4.5}$$

Let $a \in \mathbb{R}$ such that $P_{e_2^\perp}(K) = [-a,a]$, then there exists $b \geq 0$ such that $(a,b) \in K$. Set

$$S = K \cap \{s = b\} = [-a,a] \times \{b\}, \quad K_1 = \mathrm{Conv}(S,0), \quad \text{and} \quad C_1 = K_1 \cup \sigma(K_1).$$

Then, one has $C_1 \subset C$ and one easily sees that $\mathrm{Conv}(C) \setminus C \subset \mathrm{Conv}(C_1) \setminus C_1$, thus

$$\frac{\mu_m(\mathrm{Conv}(C) \setminus C)}{\mu_m(C)} \leq \frac{\mu_m(\mathrm{Conv}(C_1) \setminus C_1)}{\mu_m(C_1)}. \tag{4.6}$$

The triangle K_1 with vertices 0, (a,b) and $(-a,b)$ can be described as

$$K_1 = \{(x,s) \in \mathbb{R}^2; \frac{b}{a}|x| \leq s \leq b\}.$$

Hence, we get

$$\mu_m(C_1) = 2\mu_m(K_1) = 4 \int_0^a \int_{\frac{b}{a}x}^b s^{m-2} ds dx = \frac{4}{m} a b^{m-1}.$$

And since $\mathrm{Conv}(C_1) = [-a,a] \times [-b,b]$, we get

$$\mu_m(\mathrm{Conv}(C_1)) = 4\int_0^a \int_0^b s^{m-2} ds dx = \frac{4}{m-1} a b^{m-1}.$$

Hence

$$\frac{\mu_m(\mathrm{Conv}(C_1) \setminus C_1)}{\mu_m(C_1)} = \frac{\frac{1}{m-1} - \frac{1}{m}}{\frac{1}{m}} = \frac{1}{m-1}.$$

Using (4.6), we deduce that (4.5) holds. Moreover, there is equality in (4.6) if and only if $C = C_1$, i.e. $K = K_1$, which means that $f = \frac{1}{b} + I_{[-\frac{a}{b}, \frac{a}{b}]}$. Thus, the equality case follows. □

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